

Conductor-Rectangularized Fibers in a Primitive Möbius Tail: Tensor Large Sieve, Gauss Complementarity, and Selected Short–Short Closure

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Abstract

TPC-21 leaves the selected short–short part of a primitive Möbius residual correlation as a connected multiplicative-character moment on unit fibers modulo squarefree least common multiples. Its shared-factor formula appears to couple the two row variables through a character convolution. We prove that this convolution has an exact pairwise rectangularization. If

$$g = (u, v), \quad u = ga, \quad v = gb, \quad q = gab,$$

then, for the actual row-pair mask, character orthogonality converts the shared- g average into the compatibility condition $m \equiv n \pmod{g}$, and the remaining character coordinates form the Cartesian product $\mathfrak{X}(u) \times \mathfrak{X}(b)$. The global conductor is the product of the two coordinate conductors. This identity is applied separately to every lcm divisor pair; no common rectangular coordinate across different pairs is asserted.

For coefficients on an interval of length N , we derive from the primitive character large sieve the weighted exact-conductor estimate

$$\sum_{\substack{s \lesssim S \\ s \text{ squarefree}}} \frac{1}{\varphi(s)} \sum_{\substack{\chi \pmod{s} \\ \text{cond } \chi \asymp F}} \left| \sum_n a_n \chi(n) \right|^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F} \right) \sum_n |a_n|^2.$$

It includes conductor one and hence retains induced principal modes. Tensorization estimates an arbitrary masked coefficient matrix; no rank-one factorization or deletion of the determinant–gcd pruning is used. A second estimate exploits the exact induced Gauss factor at all additive frequencies. The Ramanujan factor $c_{q/f}(r)$ is summed against the rapidly decaying Fourier weight without imposing $(r, q) = 1$.

Writing $Q = X^{\beta+o(1)}$, $R = X^{1/2-\delta+o(1)}$, and splitting coordinate conductors at $Q^{1/2}$, the tensor-norm and direct-Gauss estimates are complementary. Every conductor block has a fixed-power saving whenever

$$1 - 4\delta < \beta < \min \left\{ \frac{2}{3}, 6\delta \right\}.$$

Subject to the inherited no-wrap and polynomial tail-mass hypotheses, either the row-occupancy range or the hybrid range above, together with the strict single-leg condition $\beta < 1/2 + \delta$, closes the connected discrepancy for the actual row-pair-only generic mask in the selected short–short packet of TPC-20–21, including the two inherited parameter profiles stated below. Combining with the Ramanujan-mean closure of TPC-21 closes that selected packet. We do not treat either large-divisor leg, reassemble the full residual-dispersion argument, prove a prime-pair asymptotic, or breach the sieve parity barrier.

Keywords: Möbius tail; CRT fibers; Dirichlet characters; exact conductors; multiplicative large sieve; induced Gauss sums; Ramanujan sums; sieve parity.

MSC 2020: Primary 11N35, 11N37; Secondary 11L05, 11L07, 11L40.

1 Introduction

After smooth separation, one-leg completion, and compression of the compatible inverse phases, TPC-20 represents a selected short-short piece of a primitive Möbius residual correlation by one finite Fourier transform on each unit fiber modulo a squarefree least common multiple [3, 4]. TPC-21 closes the constant, or Ramanujan-mean, mode and identifies the centered part exactly as a nonprincipal multiplicative-character energy [5].

That energy retains the provenance of both rows, but its initial form appears coupled. If

$$g = (u, v), \quad u = ga, \quad v = gb, \quad q = [u, v] = gab,$$

then g, a, b are pairwise coprime and TPC-21 obtains

$$\widehat{Z}_{u,v}^{\mathbf{m},\times}(\chi) = \frac{\overline{\chi(-h)}}{\varphi(g)} \sum_{\psi \in \mathfrak{X}(g)} \mathcal{B}_{\mathbf{m}}^{u,v}((\chi_g \psi) \boxtimes \chi_a, \overline{\psi} \boxtimes \chi_b). \quad (1)$$

The actual generic row-pair mask $\mathbf{m}$ cannot be discarded merely to factor $\mathcal{B}_{\mathbf{m}}$ into two Dirichlet polynomials.

The first observation here is that the convolution is exactly rectangular. Character orthogonality gives

$$\frac{1}{\varphi(g)} \sum_{\psi \bmod g} \mathcal{B}_{\mathbf{m}}^{u,v}((\chi_g \psi) \boxtimes \chi_a, \overline{\psi} \boxtimes \chi_b) = \sum_{m,n} c_{m,n} \mathbf{1}_{m \equiv n \pmod{g}} (\chi_g \boxtimes \chi_a)(m) \chi_b(n), \quad (2)$$

where $c_{m,n} = x_m y_n \mathbf{m}(m, n)$. Since $q = ub$ and $(u, b) = 1$,

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u) \times \mathfrak{X}(b), \quad (3)$$

and the two exact conductors multiply to $\text{cond } \chi$. The shared factor remains, correctly, as compatibility inside an arbitrary matrix.

The coordinates in (3) depend on (u, v) . Different lcm decompositions of the same q do not share one common rectangle. We therefore split by divisor pair first, use triangle or Cauchy over at most $3^{\omega(q)}$ pairs, and only then rectangularize. The cost is $X^{o(1)}$.

1.1 Two complementary conductor estimates

For coefficients on an interval of length N , the primitive large sieve [1, Chapter 7] gives the weighted estimate

$$\sum_{\substack{s \gtrsim S \\ s \text{ squarefree}}} \frac{1}{\varphi(s)} \sum_{\substack{\chi \pmod{s} \\ \text{cond } \chi \asymp F}} \left| \sum_n a_n \chi(n) \right|^2 \ll_{\varepsilon} X^{\varepsilon} \left(F + \frac{N}{F} \right) \sum_n |a_n|^2. \quad (4)$$

Conductor one is included, so repeated induced principal characters are not silently removed. Applying this estimate in both coordinates of (2) works for an arbitrary matrix and does not require a rank-one decomposition of the mask.

The second route uses the exact induced Gauss sum. If $f = \text{cond } \chi \mid q$, $d = q/f$, and q is squarefree, then

$$|G_q(\chi; r\overline{H})| = \begin{cases} \sqrt{f} |c_d(r)|, & (r, f) = 1, \\ 0, & (r, f) > 1. \end{cases} \quad (5)$$

No condition $(r, q) = 1$ is imposed. Rapid Fourier decay gives

$$\sum_{r \neq 0} |c_d(r)| \left| \widehat{F} \left(\frac{rJ}{Hq} \right) \right| \ll_{F, H, \varepsilon} X^\varepsilon \frac{q}{J} \tau(d), \quad (6)$$

which cancels the Poisson normalization J/q . The direct-Gauss route is strongest at low coordinate conductor, while the inherited fiber-norm route is strongest at high conductor.

1.2 The saving ledger

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad Y = X^{y+o(1)}, \quad F_i = X^{f_i+o(1)},$$

and put $k_i = \max\{f_i, \beta - f_i\}$. Relative to $Q^2 J = XQ$, the two routes save

$$\eta_A = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad (7)$$

$$\eta_B = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (8)$$

Since $f_i \leq 1/2 - \delta$ and $y \leq 1 - 2\delta$, a low-low, high-high, and mixed split gives

$$\eta_*(\beta, \delta) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{4} \right\}. \quad (9)$$

This certified lower bound is positive in the strict region

$$1 - 4\delta < \beta < \min \left\{ \frac{2}{3}, 6\delta \right\}. \quad (10)$$

It is not claimed to be the exact minimax of the two estimates.

For $\beta < 2\delta$ with a fixed margin, TPC-21 occupancy already closes every $Y \leq R^2$; equality has no top-block power margin. In the two actual inherited tail profiles, β ranges from $\log_X(LD_0)$ to $\log_X(LR^{1/2})$. The entire interval lies in (10) in the strict ranges proved below.

1.3 Scope and organization

We close the centered discrepancy for the actual row-pair-only generic mask in the selected short-short packet of TPC-20–21. With the already closed mean, this closes that selected packet. We do not treat either residual leg with a divisor exceeding R , prove the full TPC-18 residual-dispersion target, obtain a prime-pair asymptotic, or breach sieve parity.

Section 2 fixes the packet. Section 3 proves the pairwise identity. Section 4 proves (4); section 5 tensorizes it with the mask retained. Section 6 proves the all-frequency Gauss route, and section 7 completes the conductor and profile ledger. The final sections record barriers, certification, and strict non-results.

2 The inherited selected short-short interface

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and let

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad 0 < \delta < \frac{1}{2}, \quad (11)$$

with $R \geq H^2$. In one separated row packet,

$$Q = LD, \quad QJ \asymp X, \quad L > \max\{2R, 2|h|\}, \quad (12)$$

and

$$\omega = (\ell, d), \quad \ell \in [L/2, 2L] \text{ prime}, \quad d \in [D, 2D], \quad d \leq R^{1/2}, \quad m_\omega = \ell d \asymp Q. \quad (13)$$

The large prime is unique, so m_ω determines the row label. The bounded separated weights satisfy

$$\|x\|_\infty + \|y\|_\infty \ll_\varepsilon X^\varepsilon, \quad \sum_m |x_m|^2 + \sum_n |y_n|^2 \ll QX^\varepsilon. \quad (14)$$

The actual generic mask is

$$\mathbf{m}(m, n) = \mathbf{1}_{\ell_m \neq \ell_n} \mathbf{1}_{|m-n| > QX^{-\kappa_0}} \mathbf{1}_{(d_m, d_n) \leq X^{\kappa_0}}, \quad (15)$$

where $\kappa_0 > 0$ is fixed. We use only

$$|\mathbf{m}(m, n)| \leq 1, \quad \mathbf{m}(m, m) = 0, \quad \mathbf{m} \text{ is independent of } u, v, q. \quad (16)$$

Thus the theorem applies to any fixed bounded off-diagonal row-pair mask with these properties, but not as stated to a modulus-dependent mask.

The divisor coefficient is either

$$\vartheta_R(u) = \lambda'_R(u) \quad \text{or} \quad \vartheta_R(u) = b_R(u) \mathbf{1}_{u \leq R}. \quad (17)$$

Both are real, squarefree-supported on $u \leq R$, and

$$|\vartheta_R(u)| \ll_\varepsilon X^\varepsilon. \quad (18)$$

For squarefree $u, v \leq R$, put $g = (u, v)$, $q = [u, v]$, and retain only $(q, H) = 1$. A row pair is admissible when

$$(u, mH) = (v, nH) = 1, \quad m \equiv n \pmod{g}. \quad (19)$$

It determines a unique $\xi \in \mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times$ through

$$m\xi \equiv -h \pmod{u}, \quad n\xi \equiv -h \pmod{v}. \quad (20)$$

Define

$$Z_{u,v}(\xi) = \sum_{m,n} x_m y_n \mathbf{m}(m, n) \mathbf{1}_{m \equiv -h\bar{\xi} \pmod{u}} \mathbf{1}_{n \equiv -h\bar{\xi} \pmod{v}}, \quad (21)$$

$$Z_q(\xi) = \sum_{\substack{u,v \leq R \\ \text{lcm}(u,v)=q}} \vartheta_R(u) \vartheta_R(v) Z_{u,v}(\xi). \quad (22)$$

Inadmissible row-modulus combinations contribute zero.

Let

$$\bar{Z}_{u,v} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{u,v}(\xi), \quad D_{u,v} = Z_{u,v} - \bar{Z}_{u,v}, \quad (23)$$

and define $\overline{Z}_q, D_q$ similarly. Centering is linear:

$$D_q = \sum_{[u,v]=q} \vartheta_R(u) \vartheta_R(v) D_{u,v}. \quad (24)$$

This licenses pairwise triangle inequalities, not orthogonality.

We use the normalizations

$$e(t) = e^{2\pi it}, \quad e_q(t) = e(t/q), \quad c_s(r) = \sum_{\xi \in (\mathbb{Z}/s\mathbb{Z})^\times} e_s(r\xi), \quad (25)$$

and, for a function W on $\mathcal{U}_q$,

$$\|W\|_{2,\mathcal{U}_q}^2 := \sum_{\xi \in \mathcal{U}_q} |W(\xi)|^2. \quad (26)$$

For $F \in C_c^\infty(\mathbb{R})$, with $\widehat{F}(t) = \int_{\mathbb{R}} F(x) e(-tx) dx$, put

$$\mathcal{L}_q = \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) \sum_{\xi \in \mathcal{U}_q} Z_q(\xi) e_q(r\overline{H}\xi). \quad (27)$$

We write $\mathcal{L}_q^{\text{mean}}$ and $\mathcal{L}_q^{\text{disc}}$ for the same linear functional with Z_q replaced by $\overline{Z}_q$ and D_q , respectively; hence $\mathcal{L}_q = \mathcal{L}_q^{\text{mean}} + \mathcal{L}_q^{\text{disc}}$. TPC-21 proves for its mean part

$$\sum_q |\mathcal{L}_q^{\text{mean}}| \ll_\varepsilon Q^2 X^\varepsilon. \quad (28)$$

For the discrepancy, TPC-20 supplies under its no-wrap and polynomial tail-mass hypotheses

$$\sum_{q \in \mathcal{Q}} |\mathcal{L}_q^{\text{disc}}| \ll_{H,F,\varepsilon} X^{\varepsilon/2} \sqrt{\frac{J}{H}} \sum_{q \in \mathcal{Q}} \|D_q\|_{2,\mathcal{U}_q} + O_A(X^{-A} \mathcal{Z}_1). \quad (29)$$

Here $\mathcal{Z}_1$ is the total absolute coefficient mass of the separated packet. The inherited polynomial tail-mass hypothesis includes $\mathcal{Z}_1 = X^{O(1)}$, so the last term is smaller than any prescribed fixed power after choosing A . The natural scale is

$$Q^2 J \asymp XQ. \quad (30)$$

We absorb dyadic decompositions and

$$\tau(n)^C + 3^{\omega(n)} \ll_{C,\varepsilon} X^\varepsilon \quad (n \leq X^{O(1)}) \quad (31)$$

into X^ε . A conductor block of size one contains exact conductor one.

Theorem 2.1 (Selected short-short connected closure). *Assume explicitly, for some fixed $\varepsilon_0 > 0$, the TPC-20 no-wrap condition $J \geq 4HX^{\varepsilon_0}$ and its polynomial fiber-tail mass hypothesis used in (29). Write $Q = X^{\beta+o(1)}$, $J = X^{1-\beta+o(1)}$. Suppose either*

$$\beta < 2\delta \quad (32)$$

with a fixed margin, or

$$1 - 4\delta < \beta < \min\left\{\frac{2}{3}, 6\delta\right\} \quad (33)$$

with fixed margins. Assume also the strict TPC-20 single-leg condition

$$\beta < \frac{1}{2} + \delta \quad (34)$$

with a fixed margin. Then some $\eta > 0$ satisfies, uniformly for every dyadic $Y \leq R^2$,

$$\sum_{\substack{q \asymp Y \\ q \text{ squarefree} \\ (q, H) = 1}} |\mathcal{L}_q^{\text{disc}}| \ll_{F, H, \varepsilon} X Q X^{-\eta + \varepsilon}. \quad (35)$$

After the inherited dyadic and separated-packet reassembly, this and (28) close the selected short-short packet.

At $\beta = 2\delta$, occupancy has no fixed-power margin on $Y \asymp R^2$. The second alternative is the new result. Neither alternative includes the omitted large-divisor residual legs.

3 Exact pairwise conductor rectangularization

For $s \geq 1$, write

$$G_s = (\mathbb{Z}/s\mathbb{Z})^\times, \quad \mathfrak{X}(s) = \widehat{G}_s,$$

with the modulus-one groups interpreted as trivial. Characters are extended by zero away from the units, and

$$\widehat{W}^\times(\chi) = \sum_{\xi \in G_s} W(\xi) \overline{\chi(\xi)}. \quad (36)$$

Fix one squarefree pair (u, v) , and write

$$g = (u, v), \quad u = ga, \quad v = gb, \quad q = gab. \quad (37)$$

The factors g, a, b are pairwise coprime. Decompose

$$\chi = \chi_g \boxtimes \chi_a \boxtimes \chi_b \in \mathfrak{X}(q).$$

For $\rho \in \mathfrak{X}(u)$, $\sigma \in \mathfrak{X}(v)$, define

$$\mathcal{B}_{\mathfrak{m}}^{u,v}(\rho, \sigma) = \sum_{m,n} x_m y_n \mathfrak{m}(m, n) \rho(m) \sigma(n). \quad (38)$$

Theorem 3.1 (Pairwise rectangularization). *Put*

$$c_{m,n}^{(g)} := x_m y_n \mathfrak{m}(m, n) \mathbf{1}_{m \equiv n \pmod{g}}. \quad (39)$$

Then

$$\begin{aligned} \widehat{Z}_{u,v}^\times(\chi) &= \frac{\overline{\chi(-h)}}{\varphi(g)} \sum_{\psi \in \mathfrak{X}(g)} \mathcal{B}_{\mathfrak{m}}^{u,v}((\chi_g \psi) \boxtimes \chi_a, \overline{\psi} \boxtimes \chi_b) \\ &= \overline{\chi(-h)} \sum_{m,n} c_{m,n}^{(g)} (\chi_g \boxtimes \chi_a)(m) \chi_b(n). \end{aligned} \quad (40)$$

There is no conjugation of the n -character in the last line and no remaining factor $1/\varphi(g)$.

Proof. The first equality is the exact shared-factor transform of TPC-21. Expanding its right side gives

$$\sum_{m,n} x_m y_n \mathfrak{m}(m, n) \chi_g(m) \chi_a(m) \chi_b(n) \frac{1}{\varphi(g)} \sum_{\psi \in \mathfrak{X}(g)} \psi(m) \overline{\psi(n)}.$$

For $m, n \in G_g$, the normalized inner sum is $\mathbf{1}_{m \equiv n \pmod{g}}$. If $m \notin G_g$, then $\chi_g(m) = 0$. If $m \in G_g$ and $m \equiv n \pmod{g}$, then $n \in G_g$ automatically. The zero-extended formula is therefore exactly (40). $\square$

Put

$$\alpha = \chi_g \boxtimes \chi_a \in \mathfrak{X}(u), \quad \beta = \chi_b \in \mathfrak{X}(b).$$

Corollary 3.2 (Product coordinates and conductors). *The map $\chi \mapsto (\alpha, \beta)$ is a bijection*

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u) \times \mathfrak{X}(b). \quad (41)$$

If $f_1 = \text{cond } \alpha$, $f_2 = \text{cond } \beta$, then

$$\text{cond } \chi = f_1 f_2. \quad (42)$$

The global principal character corresponds exactly to $(\mathbf{1}_u, \mathbf{1}_b)$.

Proof. Since $q = ub$ and $(u, b) = 1$, CRT gives $G_q = G_u \times G_b$, proving the bijection. Exact conductors on coprime factors multiply. The principal assertion follows locally. $\square$

There is a symmetric coordinate choice

$$\mathfrak{X}(q) \simeq \mathfrak{X}(a) \times \mathfrak{X}(v),$$

obtained by assigning the shared g -character to the second row. We use (41).

3.1 Why pairwise must not be replaced by global

For squarefree q , every prime divisor can occur in the first modulus only, the second only, or both. Thus

$$\#\{(u, v) : [u, v] = q\} = 3^{\omega(q)}. \quad (43)$$

Restrictions $u, v \leq R$ only reduce this count. By (24), either Minkowski in the fiber norm or Cauchy in character energy costs only $X^{o(1)}$.

The coordinate $\mathfrak{X}(u) \times \mathfrak{X}(b)$ changes with the divisor pair. We first split the scalar or norm into pair contributions, then split exact conductors in that pair's coordinates, and only afterward collect dyadic conductor sizes.

3.2 The shared-factor coefficient energy

Proposition 3.3 (Total compatibility-matrix energy). *Under (14) and (16),*

$$\boxed{\sum_{g \leq R} \sum_{m, n} |c_{m, n}^{(g)}|^2 \ll_{\varepsilon} Q^2 X^{\varepsilon}.} \quad (44)$$

The estimate remains true after imposing squarefreeness, $(g, H) = 1$, or any inherited divisor-pair restriction.

Proof. The mask forces $m \neq n$. Therefore

$$\begin{aligned} \sum_{g \leq R} \sum_{m, n} |c_{m, n}^{(g)}|^2 &\leq \sum_{m \neq n} |x_m|^2 |y_n|^2 \sum_{g|m-n} 1 \\ &\leq \max_{0 < |t| \ll Q} \tau(|t|) \left(\sum_m |x_m|^2 \right) \left(\sum_n |y_n|^2 \right). \end{aligned}$$

Use (14) and the divisor bound. Every omitted restriction removes nonnegative summands. $\square$

If the diagonal $m = n$ were restored, every $g \leq R$ would divide the zero difference. The divisor-count proof above would then acquire an additional diagonal term that must be estimated separately. In the present bounded-weight interface that term is still acceptable, since

$$R \sum_m |x_m y_m|^2 \leq R \|x\|_\infty^2 \|y\|_2^2 \ll_\varepsilon R Q X^\varepsilon \ll_\varepsilon Q^2 X^\varepsilon,$$

where $R < Q$ follows from (12) and nonempty integer row support. Thus retaining the actual mask is essential for the exact packet identity, but no false claim that the diagonal alone breaks the displayed scale is needed.

4 A weighted exact-conductor large sieve

The primitive multiplicative large sieve states that, for coefficients on an interval of length N ,

$$\sum_{f \leq T} \frac{f}{\varphi(f)} \sum_{\chi^* \pmod{f}}^* \left| \sum_n a_n \chi^*(n) \right|^2 \ll (T^2 + N) \sum_n |a_n|^2 \quad (45)$$

[1, Chapter 7]. We need a normalization that keeps the exact conductor while the ambient modulus varies.

Theorem 4.1 (Weighted exact-conductor large sieve). *Let a_n be supported on an interval of length at most N . Let $s \asymp S$ run over squarefree moduli, and let $\text{cond } \chi \asymp F$ denote a dyadic exact-conductor block, with the block $F = 1$ containing only conductor one. If all parameters are at most $X^{O(1)}$, then*

$$\boxed{\sum_{\substack{s \asymp S \\ s \text{ squarefree}}} \frac{1}{\varphi(s)} \sum_{\substack{\chi \in \mathfrak{X}(s) \\ \text{cond } \chi \asymp F}} \left| \sum_n a_n \chi(n) \right|^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F} \right) \sum_n |a_n|^2.} \quad (46)$$

The implied constant is uniform in the location of the coefficient interval and in S .

Proof. First suppose $F > 1$. If $\chi \pmod{s}$ has exact conductor f , then $s = fd$, $(f, d) = 1$, and the squarefree hypothesis makes the factorization unique. For the inducing primitive character $\chi^* \pmod{f}$,

$$\chi(n) = \chi^*(n) \mathbf{1}_{(n, d)=1}. \quad (47)$$

For fixed d , set $a_n^{(d)} = a_n \mathbf{1}_{(n, d)=1}$. The conditions $s \asymp S$, $f \asymp F$ confine d to $d \asymp S/F$, up to a bounded dyadic refinement. Moreover

$$\frac{1}{\varphi(fd)} = \frac{1}{fd} \frac{f}{\varphi(f)} \frac{d}{\varphi(d)} \ll_\varepsilon \frac{X^\varepsilon}{S} \frac{f}{\varphi(f)}. \quad (48)$$

Dropping coprimality and range restrictions only enlarges the nonnegative sum. Applying (45) in f for each d gives

$$\frac{X^\varepsilon}{S} (F^2 + N) \sum_n |a_n^{(d)}|^2 \leq \frac{X^\varepsilon}{S} (F^2 + N) \sum_n |a_n|^2.$$

There are $O(S/F)$ possible d , so the total is

$$\ll X^\varepsilon \frac{F^2 + N}{F} \sum_n |a_n|^2,$$

which is (46).

For $F = 1$, the only character modulo s is the principal character induced from modulus one. By Cauchy–Schwarz,

$$\left| \sum_{(n,s)=1} a_n \right|^2 \leq N \sum_n |a_n|^2.$$

The standard dyadic estimate $\sum_{s \asymp S} 1/\varphi(s) \ll_\varepsilon X^\varepsilon$ then gives

$$\ll X^\varepsilon N \sum_n |a_n|^2,$$

which is bounded by the right side of (46) with $F = 1$. $\square$

Remark 4.2 (Why conductor one matters). A fixed primitive character is repeated on every ambient modulus to which it induces, with an additional coprimality cutoff as in (47). Thus a principle treating all nonprincipal characters as primitive characters of size S is false. The factor $F + N/F$ retains low-conductor repetition. In the two-dimensional application, blocks with $F_1 = 1 < F_2$ or $F_2 = 1 < F_1$ are genuine connected modes and must not be deleted.

The theorem also has a Hilbert-valued form. If a_n takes values in a finite-dimensional Hilbert space $\mathcal{H}$, then

$$\sum_{s \asymp S} \frac{1}{\varphi(s)} \sum_{\text{cond } \chi \asymp F} \left\| \sum_n a_n \chi(n) \right\|_{\mathcal{H}}^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F} \right) \sum_n \|a_n\|_{\mathcal{H}}^2. \quad (49)$$

Expand in an orthonormal basis, apply theorem 4.1 coordinatewise, and sum.

5 Tensor conductor energy with the mask retained

For $u = ga$, $v = gb$, set

$$C_{g,a,b}(\alpha, \beta) := \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \beta(n), \quad \alpha \in \mathfrak{X}(u), \quad \beta \in \mathfrak{X}(b). \quad (50)$$

By theorem 3.1, this is the full masked pair transform after removal of the harmless phase $\overline{\chi(-h)}$.

Put

$$\mathcal{K}_Q(F) := F + \frac{Q}{F}. \quad (51)$$

For dyadic coordinate-conductor sizes F_1, F_2 , define the pair-separated energy

$$\begin{aligned} \mathcal{E}_Y(F_1, F_2) := & \sum_{\substack{g,a,b \text{ pairwise coprime} \\ ga, gb \leq R, \text{ } gab \asymp Y \\ gab \text{ squarefree, } (gab, H)=1}} \frac{|\vartheta_R(ga) \vartheta_R(gb)|^2}{\varphi(gab)} \\ & \times \sum_{\substack{\alpha \in \mathfrak{X}(ga) \\ \text{cond } \alpha \asymp F_1}} \sum_{\substack{\beta \in \mathfrak{X}(b) \\ \text{cond } \beta \asymp F_2}} |C_{g,a,b}(\alpha, \beta)|^2. \end{aligned} \quad (52)$$

This is intentionally a sum of pair energies, not the energy of a fictional global rectangular projection of Z_q .

Theorem 5.1 (Arbitrary-matrix tensor conductor bound). *Uniformly for $Y \leq R^2$ and $1 \leq F_i \leq R$,*

$$\boxed{\mathcal{E}_Y(F_1, F_2) \ll_\varepsilon X^\varepsilon Q^2 \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (53)$$

No separability assumption on $c_{m,n}^{(g)}$ is made.

Proof. Use (18), split $u = ga$ and b into dyadic ranges, and fix g . Since $q = gab = ub$ and $(u, b) = 1$, $\varphi(q) = \varphi(u)\varphi(b)$. Retaining squarefreeness but removing the conditions

$$g \mid u, \quad (u, b) = 1, \quad ub \asymp Y, \quad gb \leq R$$

enlarges the nonnegative sum. It is therefore enough to bound

$$\sum_{\substack{u \asymp U \\ u \text{ squarefree}}} \frac{1}{\varphi(u)} \sum_{\substack{\alpha \in \mathfrak{X}(u) \\ \text{cond } \alpha \asymp F_1}} \sum_{\substack{b \asymp B \\ b \text{ squarefree}}} \frac{1}{\varphi(b)} \sum_{\substack{\beta \in \mathfrak{X}(b) \\ \text{cond } \beta \asymp F_2}} \left| \sum_{m, n} c_{m, n}^{(g)} \alpha(m) \beta(n) \right|^2. \quad (54)$$

Fix b, β and apply [theorem 4.1](#) in u, α to

$$a_m(\beta) = \sum_n c_{m, n}^{(g)} \beta(n).$$

This contributes $\mathcal{K}_Q(F_1)$ and leaves

$$\mathcal{K}_Q(F_1) \sum_{\substack{b \asymp B \\ b \text{ squarefree}}} \frac{1}{\varphi(b)} \sum_{\text{cond } \beta \asymp F_2} \sum_m \left| \sum_n c_{m, n}^{(g)} \beta(n) \right|^2.$$

Apply the same theorem in b, β , separately for each m . The result is

$$\ll_{\varepsilon} X^{\varepsilon} \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2) \sum_{m, n} |c_{m, n}^{(g)}|^2. \quad (55)$$

This is a composition of two Hilbert-space operator bounds, so it does not factor the coefficient matrix.

Sum (55) over g and the logarithmically many dyadic u, b -ranges. Then [proposition 3.3](#) proves (53). $\square$

5.1 The inherited norm route

For a fixed pair, multiplicative Parseval and [corollary 3.2](#) decompose its centered fiber into the blocks $(F_1, F_2) \neq (1, 1)$. Let $D_{u, v}^{F_1, F_2}$ denote the inverse character transform of one such block, with the factor $\vartheta_R(u)\vartheta_R(v)$ included. Then

$$\|D_{u, v}^{F_1, F_2}\|_{2, \mathcal{U}_q}^2 = \frac{|\vartheta_R(u)\vartheta_R(v)|^2}{\varphi(q)} \sum_{\substack{\text{cond } \alpha \asymp F_1 \\ \text{cond } \beta \asymp F_2}} |C_{g, a, b}(\alpha, \beta)|^2. \quad (56)$$

The phase $\overline{\chi(-h)}$ has modulus one.

Let $\mathcal{L}_{u, v}^{F_1, F_2}$ be the scalar packet (27) formed from this pair block, and define the pair-separated dyadic aggregate

$$\mathfrak{L}_Y(F_1, F_2) := \sum_{\substack{g, a, b \text{ pairwise coprime} \\ ga, gb \leq R, \text{ } gab \asymp Y \\ gab \text{ squarefree, } (gab, H)=1}} |\mathcal{L}_{ga, gb}^{F_1, F_2}|. \quad (57)$$

This is one object. The subscripts A and B used informally below name two estimates for it, not two different projections. By linearity, pairwise triangle inequality, and dyadic conductor decomposition,

$$\sum_{q \asymp Y} |\mathcal{L}_q^{\text{disc}}| \leq \sum_{\substack{F_1, F_2 \\ (F_1, F_2) \neq (1, 1)}} \mathfrak{L}_Y(F_1, F_2). \quad (58)$$

The number of triples (g, a, b) with $gab \asymp Y$ is $\ll YX^\varepsilon$, by (43). Cauchy–Schwarz and theorem 5.1 therefore give

$$\sum_{\substack{q \asymp Y \\ \text{lcm}(u,v)=q}} \|D_{u,v}^{F_1, F_2}\|_{2, \mathcal{U}_q} \ll_\varepsilon X^\varepsilon Q \sqrt{Y \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (59)$$

The notation on the left means the sum over the pair contributions before their lcm reassembly.

Corollary 5.2 (Tensor-norm scalar bound). *Under the inherited hypotheses of (29), every nonprincipal coordinate-conductor block satisfies*

$$\boxed{\mathfrak{L}_Y(F_1, F_2) \ll_\varepsilon X^\varepsilon Q \sqrt{Y \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}}. \quad (60)$$

Proof. Use (24) and Minkowski before (29), then insert (59). The Fourier-tail error is already part of the inherited polynomial-mass hypothesis. $\square$

The pairwise triangle inequality may discard cancellation between lcm decompositions, but it costs no fixed power and is sufficient for the claimed closure. The next section proves a second estimate for this same aggregate.

6 The direct Gauss–Ramanujan route

The norm route does not use the conductor sensitivity of the additive Gauss factor. We now exploit it before taking a fiber norm. For $\chi \in \mathfrak{X}(q)$, define

$$G_q(\chi; t) = \sum_{\xi \in G_q} \chi(\xi) e_q(t\xi). \quad (61)$$

Lemma 6.1 (Induced Gauss factor at every frequency). *Let q be squarefree, let $f = \text{cond } \chi \mid q$, let $\chi^* \bmod f$ induce χ , and put $d = q/f$. Then*

$$G_q(\chi; t) = \begin{cases} \overline{\chi^*(t)} \chi^*(d) \tau_f(\chi^*) c_d(t), & (t, f) = 1, \\ 0, & (t, f) > 1. \end{cases} \quad (62)$$

Here $\tau_1(\mathbf{1}) = 1$. Consequently, for $f > 1$,

$$|G_q(\chi; t)| = \sqrt{f} |c_d(t)| \mathbf{1}_{(t, f)=1}. \quad (63)$$

Proof. CRT factors (61) into a primitive Gauss sum modulo f and a Ramanujan sum modulo d . The first factor vanishes when $(t, f) > 1$; otherwise the usual change of variables gives $\overline{\chi^*(t)} \chi^*(d) \tau_f(\chi^*)$. The second factor is $c_d(t)$. Primitive Gauss magnitude is $\sqrt{f}$. $\square$

Since $(H, q) = 1$, replacing t by $r\overline{H}$ does not change the condition $(r, f) = 1$, and it changes only a unit-modulus character phase.

Lemma 6.2 (Ramanujan-weighted nonzero frequencies). *If s is squarefree, $K > 0$, and F is fixed and smooth with rapidly decaying Fourier transform, then*

$$\boxed{\sum_{r \neq 0} |c_s(r)| |\widehat{F}(r/K)| \ll_F K \tau(s)}. \quad (64)$$

The exclusion of $r = 0$ is essential.

Proof. For squarefree s ,

$$|c_s(r)| = \varphi((s, r)) \leq (s, r) = \sum_{e|(s, r)} \varphi(e). \quad (65)$$

For every $e \geq 1$, rapid decay gives uniformly in $K > 0$

$$\sum_{\substack{r \neq 0 \\ e|r}} |\widehat{F}(r/K)| = \sum_{k \neq 0} |\widehat{F}(ke/K)| \ll_F \frac{K}{e}. \quad (66)$$

When $e > K$, take two or more powers of decay; this is precisely where removing $r = 0$ matters. Insert (65), interchange the divisor and frequency sums, and use $\sum_{e|s} \varphi(e)/e \leq \tau(s)$. $\square$

6.1 One exact pair-conductor block

Fix g, a, b , let $u = ga$, $v = gb$, $q = gab$, and fix exact coordinate conductors

$$f_1 = \text{cond } \alpha, \quad f_2 = \text{cond } \beta, \quad f = f_1 f_2, \quad d = \frac{q}{f}. \quad (67)$$

Define

$$A_{g,a,b;f_1,f_2}^2 := |\vartheta_R(u)\vartheta_R(v)|^2 \sum_{\substack{\alpha \in \mathfrak{X}(u), \text{ cond } \alpha = f_1 \\ \beta \in \mathfrak{X}(b), \text{ cond } \beta = f_2}} |C_{g,a,b}(\alpha, \beta)|^2. \quad (68)$$

By (42), all characters in this block have global conductor f . Let $D_{g,a,b;f_1,f_2}$ be its inverse multiplicative-character transform, including the factor $\vartheta_R(u)\vartheta_R(v)$ and the unit-modulus phase $\chi(-h)$, and define

$$S_{g,a,b;f_1,f_2}(r) := \sum_{\xi \in \mathcal{U}_q} D_{g,a,b;f_1,f_2}(\xi) e_q(r \overline{H} \xi). \quad (69)$$

Multiplicative inversion and lemma 6.1 show that

$$|S_{g,a,b;f_1,f_2}(r)| \leq \frac{f}{\varphi(q)} |c_d(r)| A_{g,a,b;f_1,f_2}. \quad (70)$$

Indeed, the number of primitive characters of conductor f is at most f ; Cauchy-Schwarz combines its square root with the Gauss magnitude $\sqrt{f}$ to give exactly f . Terms with $(r, f) > 1$ vanish and may be retained harmlessly on the right.

Insert (70) into (27). Since $(H, d) = 1$, multiplicativity gives $|c_H(r)c_d(r)| = |c_{Hd}(r)|$. Apply lemma 6.2 with $s = Hd$ and $K = Hq/J$. The factors $J/(Hq)$ and Hq/J cancel, yielding

$$\begin{aligned} |\mathcal{L}_{g,a,b;f_1,f_2}| &\ll_{F,H} \frac{f}{\varphi(q)} \tau(Hd) A_{g,a,b;f_1,f_2} \\ &\ll_{F,H,\varepsilon} X^\varepsilon \frac{f}{q} A_{g,a,b;f_1,f_2}. \end{aligned} \quad (71)$$

The second line uses $q/\varphi(q) \ll_\varepsilon X^\varepsilon$. This estimate uses all nonzero frequencies, not merely those coprime to q .

6.2 Dyadic conductor aggregation

Exact conductors must be split before dyadic aggregation: the Ramanujan factor $c_{q/(f_1 f_2)}(r)$ changes with the exact pair (f_1, f_2) . The number of exact conductor pairs and lcm divisor pairs associated with a fixed squarefree q is $\tau(q)^{O(1)}$, hence $X^{o(1)}$.

Sum (71) over $f_i \asymp F_i$, over divisor pairs, and over $q \asymp Y$. Weighted Cauchy–Schwarz gives

$$\sum \frac{A_{g,a,b,f_1,f_2}}{q} \leq \left(\sum \frac{A_{g,a,b,f_1,f_2}^2}{\varphi(q)} \right)^{1/2} \left(\sum \frac{\varphi(q)}{q^2} \right)^{1/2} \ll_\varepsilon X^\varepsilon \mathcal{E}_Y(F_1, F_2)^{1/2}. \quad (72)$$

More explicitly, if $I = (g, a, b, f_1, f_2)$ is the full summation index, then

$$\sum_I \frac{\varphi(q_I)}{q_I^2} \ll X^\varepsilon \sum_{q \asymp Y} \frac{\varphi(q)}{q^2} \ll X^\varepsilon.$$

There is no factor $\sqrt{Y}$.

Theorem 6.3 (Direct-Gauss conductor bound). *Every nonprincipal dyadic coordinate-conductor block satisfies*

$$\boxed{\mathfrak{L}_Y(F_1, F_2) \ll_\varepsilon X^\varepsilon Q F_1 F_2 \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}}. \quad (73)$$

Proof. Equations (71) and (72) give

$$\mathfrak{L}_Y(F_1, F_2) \ll_\varepsilon X^\varepsilon F_1 F_2 \mathcal{E}_Y(F_1, F_2)^{1/2}.$$

Insert (53). □

Combining (60) and (73) for the same object gives the compact cell estimate

$$\boxed{\mathfrak{L}_Y(F_1, F_2) \ll_\varepsilon X^\varepsilon Q \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}}. \quad (74)$$

The block $(F_1, F_2) = (1, 1)$ is the global principal character and belongs to the mean already handled in (28). The one-sided blocks $(1, F_2)$ and $(F_1, 1)$ with the other conductor nontrivial remain in [theorem 6.3](#).

7 Conductor partition and exponent closure

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad Y = X^{y+o(1)}, \quad F_i = X^{f_i+o(1)}. \quad (75)$$

Since $F_i \leq R$,

$$0 \leq f_i \leq r_0 := \frac{1}{2} - \delta, \quad y \leq 1 - 2\delta. \quad (76)$$

Set

$$k_i := \max\{f_i, \beta - f_i\}. \quad (77)$$

Then $\mathcal{K}_Q(F_i) = X^{k_i+o(1)}$.

The natural scale (30) has exponent $1 + \beta$. The tensor-norm bound (60) has exponent

$$\beta + \frac{1 - \beta + y + k_1 + k_2}{2},$$

and the direct bound (73) has exponent

$$\beta + f_1 + f_2 + \frac{k_1 + k_2}{2}.$$

Thus their saving exponents are

$$\eta_A = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad \eta_B = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (78)$$

For each block we use the larger of these two savings.

Proposition 7.1 (Three-cell conductor ledger). *Under (76), every conductor pair satisfies*

$$\max\{\eta_A, \eta_B\} \geq \eta_*(\beta, \delta), \quad (79)$$

where

$$\eta_*(\beta, \delta) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{4} \right\}. \quad (80)$$

Proof. It suffices to take the worst denominator exponent $y = 1 - 2\delta$, since decreasing y improves η_A .

If $f_1, f_2 \leq \beta/2$, then

$$k_1 + k_2 = 2\beta - f_1 - f_2.$$

Consequently

$$\eta_B = 1 - \beta - \frac{f_1 + f_2}{2} \geq 1 - \frac{3\beta}{2}.$$

If $f_1, f_2 \geq \beta/2$, then

$$k_1 + k_2 = f_1 + f_2 \leq 1 - 2\delta.$$

Hence

$$\eta_A \geq \frac{\beta + 4\delta - 1}{2}.$$

It remains to consider a mixed block. Write

$$x \leq \frac{\beta}{2} \leq z, \quad d = z - x \geq 0.$$

Then $k_1 + k_2 = \beta + d$, after interchanging the coordinates if necessary, and

$$\eta_A \geq \delta - \frac{d}{2}. \quad (81)$$

Since $z \leq 1/2 - \delta$, direct substitution in η_B gives

$$\eta_B = 1 - \frac{\beta}{2} - \frac{x}{2} - \frac{3z}{2} \geq 2\delta + \frac{d}{2} - \frac{\beta}{2}. \quad (82)$$

The maximum is at least the average of the right sides of (81) and (82), namely $(6\delta - \beta)/4$. Combining the three cases proves (80). $\square$

Corollary 7.2 (Hybrid closure region). *Assume the inherited no-wrap and polynomial tail-mass hypotheses used in (29). If*

$$1 - 4\delta < \beta < \min \left\{ \frac{2}{3}, 6\delta \right\} \quad (83)$$

with fixed margins, then $\eta_(\beta, \delta) > 0$ with a fixed margin. Summing all dyadic conductor blocks other than $(1, 1)$ proves (35).*

Proof. The three strict inequalities in (83) are exactly the positivity conditions for the three entries in (80). There are $O((\log X)^2)$ coordinate-conductor blocks. Exact-conductor and divisor-pair refinements cost $X^{o(1)}$, so choose ε smaller than the fixed saving margin. The principal block is the mean, not the discrepancy. $\square$

7.1 The occupancy alternative and its gap

TPC-21 occupancy gives on $q \asymp Y$

$$\sum_q \|D_q\|_2 \ll_\varepsilon Q^2 \sqrt{Y} X^\varepsilon.$$

After (29), its ratio to $Q^2 J$ is $X^{(y-(1-\beta))/2+o(1)}$. At the largest $y = 1 - 2\delta$, the saving is

$$\eta_{\text{occ}} = \frac{2\delta - \beta}{2}. \quad (84)$$

Thus occupancy closes every $Y \leq R^2$ only when $\beta < 2\delta$ with a fixed margin. At equality the top block has zero margin. If $\delta < 1/6$, the interval between 2δ and $1 - 4\delta$ is a genuine gap between the two sufficient routes. The actual profiles below avoid it; no general claim that the gap is empty is made.

Proof of theorem 2.1. In the occupancy alternative, (84) and the fixed margin in $\beta < 2\delta$ give a uniform positive saving for every $Y \leq R^2$. In the hybrid alternative, corollary 7.2 gives the uniform positive saving η_* . In either case, choose η below the relevant fixed margin and absorb the logarithmically many dyadic blocks into X^ε . The no-wrap hypothesis licenses (29); since $Z_1 = X^{O(1)}$, its tail is absorbed by taking A sufficiently large. Finally the strict single-leg condition licenses the inherited separated-packet reassembly. This proves (35); combining it with (28) closes exactly the selected packet claimed in the theorem. $\square$

7.2 The two inherited tail profiles

The TPC-18 tail has $D > D_0$, not $D \geq 1$. Hence a profile is a whole interval of row exponents. We record the actual endpoints from the TPC-17 profile ledger [2].

Corollary 7.3 (Published tail profile). *Assume the no-wrap and polynomial tail-mass hypotheses of theorem 2.1. Let*

$$L = X^{10/21-\sigma}, \quad D_0 = X^{1/21-\sigma}, \quad D_0 < D \leq R^{1/2}. \quad (85)$$

Then

$$\frac{11}{21} - 2\sigma < \beta \leq \frac{61}{84} - \sigma - \frac{\delta}{2} + o(1). \quad (86)$$

Write

$$\delta = \left(\frac{19}{126} - \frac{2}{3}\sigma \right) + t, \quad t > 0. \quad (87)$$

Assume also the inherited upper bound $\delta < 1/4$. Uniformly over (86), the three entries in the ledger are bounded below by

$$\frac{1}{42} + \sigma + \frac{3t}{4}, \quad \frac{4}{63} - \frac{7\sigma}{3} + 2t, \quad \frac{4}{63} - \frac{5\sigma}{6} + \frac{13t}{8}. \quad (88)$$

In particular, the inherited range $0 < \sigma < 1/100$ has a fixed positive conductor margin for every fixed $t > 0$, and theorem 2.1 applies to every tail slice.

Proof. The lower endpoint is $\log_X(LD_0)$; the upper endpoint is $\log_X(LR^{1/2})$. The low-low and mixed expressions in (80) are worst at the upper endpoint, while the high-high expression is worst at the lower endpoint. Substitute (87). Moreover,

$$\left(\frac{1}{2} + \delta \right) - \beta_{\max} = \frac{3t}{2},$$

so the strict single-leg margin is automatic. The inherited bound $1/100 < 4/147$ makes all three expressions positive even as $t \downarrow 0$. $\square$

Corollary 7.4 (Version-locked Li-v6 tail profile). *Assume the no-wrap and polynomial tail-mass hypotheses of [theorem 2.1](#). Let*

$$L = X^{8/17-\sigma}, \quad D_0 = X^{1/17+\sigma/2}, \quad D_0 < D \leq R^{1/2}. \quad (89)$$

Then

$$\frac{9}{17} - \frac{\sigma}{2} < \beta \leq \frac{49}{68} - \sigma - \frac{\delta}{2} + o(1). \quad (90)$$

Write

$$\delta = \left(\frac{5}{34} - \frac{2}{3}\sigma \right) + t, \quad t > 0. \quad (91)$$

Assume also the inherited upper bound $\delta < 1/4$. Uniformly over [\(90\)](#), the three ledger entries are bounded below by

$$\frac{1}{34} + \sigma + \frac{3t}{4}, \quad \frac{1}{17} - \frac{19\sigma}{12} + 2t, \quad \frac{1}{17} - \frac{5\sigma}{6} + \frac{13t}{8}. \quad (92)$$

Thus the inherited range $0 < \sigma < 1/1000$ has a fixed positive conductor margin for every fixed $t > 0$, and [theorem 2.1](#) applies to every tail slice.

Proof. As above, use the lower endpoint for the high-high entry and the upper endpoint for the other two. Substitute [\(91\)](#). Again,

$$\left(\frac{1}{2} + \delta \right) - \beta_{\max} = \frac{3t}{2}.$$

Here $1/1000 < 12/323$. □

At $\sigma = 0$ and $t \downarrow 0$, the published ledger has endpoint margins

$$\left(\frac{1}{42}, \frac{4}{63}, \frac{4}{63} \right), \quad (93)$$

and the Li ledger has

$$\left(\frac{1}{34}, \frac{1}{17}, \frac{1}{17} \right). \quad (94)$$

More generally, $4/147$ and $12/323$ are the respective arithmetic thresholds for positivity uniformly as $t \downarrow 0$. They are not absolute barriers: a larger fixed t can compensate a larger σ , subject to the remaining inherited constraints. Nor do these wider arithmetic thresholds extend the upstream profile theorem; the applications above use its original, stricter σ -ranges.

8 Barriers, order of operations, and strict scope

The positive theorem depends on several exact choices. This section separates them from tempting but invalid shortcuts.

8.1 A low-conductor repetition no-go

The exact-conductor decomposition is not optional. Set $S \asymp N^{1/2}$, fix a nonprincipal primitive character $\chi_0 \bmod f_0$ of fixed squarefree conductor $f_0 > 1$, and induce it to squarefree moduli $s = f_0 d \asymp S$. Take

$$a_n = \overline{\chi_0(n)} \mathbf{1}_{1 \leq n \leq N}.$$

Restrict d to primes in a fixed dyadic interval, excluding the finitely many divisors of f_0 . The prime number theorem and direct inclusion–exclusion give

$$\left| \sum_n a_n \chi_s(n) \right| = \sum_{\substack{n \leq N \\ (n, f_0 d) = 1}} 1 = N \frac{\varphi(f_0 d)}{f_0 d} + O(\tau(f_0 d)) \gg_{f_0} N.$$

Consequently the weighted sum over these $d \asymp S/f_0$ has size

$$\gg \frac{N^2}{f_0 \log S}. \quad (95)$$

An estimate treating these characters as primitive at ambient scale would have only $N^{3/2+o(1)}$ on the right. This is an operator obstruction to the false large sieve, not a lower bound for the actual TPC scalar packet.

8.2 Four orders that cannot be interchanged

- (i) *Divisor pair before rectangular coordinates.* Different (u, v) with the same lcm use different products $\mathfrak{X}(u) \times \mathfrak{X}(v/(u, v))$. A common global (F_1, F_2) -coordinate is not defined.
- (ii) *Exact conductor before the Ramanujan frequency sum.* Inside one dyadic block, $c_{q/(f_1 f_2)}(r)$ still depends on the exact conductors. We first apply (71) at exact f_1, f_2 , then pay the divisor-soft dyadic reassembly.
- (iii) *Remove zero frequency before lemma 6.2.* If $r = 0$ were included, $c_s(0) = \varphi(s)$ would destroy (66) when $e > K$. The selected Poisson packet already has $r \neq 0$.
- (iv) *Keep the mask before nonlinear energy.* Estimate (44) uses the actual off-diagonal support. Deleting the mask before squaring would restore $m = n$, for which every shared factor divides the difference.

8.3 Why this is not a projective-mask theorem

Some mask components admit inexpensive separated representations, but the divisor-gcd cutoff does not automatically have a sufficiently small naive projective norm. The tensor proof avoids the issue: $c_{m,n}^{(g)}$ is arbitrary. It does require the same matrix while the output moduli vary for fixed g . A mask depending on u, v or q is therefore not covered without a new argument.

The large shared-factor and moment-diagonal bounds obtainable directly from TPC-21 remain useful cross-checks, but they are not needed to manufacture the saving here. The power saving comes from product coordinates, the weighted conductor large sieve, and the complementary induced-Gauss estimate.

8.4 Closed and open objects

Object	Status after TPC-22
Ramanujan mean of the selected short-short packet	Closed in TPC-21 and imported through (28).
Connected discrepancy of the same packet, with the actual row-pair-only generic mask	Closed in the parameter regions of theorem 2.1.
One-sided residual terms with $u > R$ or $v > R$	Not represented by (22); open.
Uniform restoration of all earlier normal-form pieces and endpoint terms in the full TPC-18 residual	Not proved by the selected-packet estimate; open.
Hardy–Littlewood asymptotic, twin-prime lower bound, or parity breach	Not claimed and not implied.

The phrase *selected short-short closure* always refers to the first two rows of this table. It is not an abbreviation for the full residual-dispersion theorem.

8.5 Parity

The character tensor supplies cancellation in a previously unresolved analytic norm. It does not alter the sign-insensitivity of the underlying upper-bound sieve and does not produce a positive lower bound for prime pairs. Nothing here distinguishes an actual prime-pair main term from every parity-compatible surrogate at the final sieve stage. The theorem is a rigorous advance inside the TPC normal form, not a claim to have crossed the classical parity barrier.

9 Exact finite certificate and conclusion

The proofs above are symbolic. A deterministic finite certificate is included to audit the identities most vulnerable to normalization or coordinate errors. It is not numerical evidence for an unproved asymptotic and is not used to replace any analytic estimate.

9.1 Certificate design

The script

`experiments/tpc22_certificate.py`

uses only the Python standard library. Roots of unity are represented exactly in $\mathbf{F}_{61}$, whose multiplicative group contains the orders needed for the squarefree test moduli. Rational exponent calculations use `fractions.Fraction`.

The primary model is

$$q = 30, \quad (u, v) = (10, 15), \quad (g, a, b) = (5, 2, 3),$$

together with the reversed divisor pair. The certificate checks:

- (1) the shared- g convolution and (40) character by character;
- (2) both product-coordinate choices and exact conductor multiplication;

- (3) the identification of the global principal character with the joint principal coordinate, while retaining one-sided principal modes;
- (4) the induced-character Gauss formula at every residue frequency, including frequencies not coprime to q ;
- (5) the exact rational low–low, high–high, and mixed ledger;
- (6) the two actual-tail profile endpoint intervals, gains, and arithmetic endpoint-stability thresholds.

Hard-failure variants deliberately omit compatibility, replace $b = v/g$ by v , omit the shared g -character, retain an extra $1/\varphi(g)$, or delete a one-sided principal block. Each altered formula is required to disagree with the exact model. The ordinary and `python -0` runs must produce byte-identical JSON, so correctness does not depend on Python assertions. All flags for a twin-prime result, Hardy–Littlewood asymptotic, full residual closure, or parity breakthrough are set to false.

The archived run performs 346 exact checks. Its JSON and source SHA-256 hashes are, respectively,

```
772aa4a68009d2ffb7960f2d27db287e52c7a80aa7cd915293e4fc8dea3fa934
2cc5e3fcdab3a51950f60c726ba8d64a96542c5fde115ac32545834be54e2964
```

The certificate digest is

```
c93ff8e6755ab0db536621e3b2c2b3f8332950da0a5784f60138912a2159fab8.
```

9.2 Conclusion

The shared-factor character convolution left by TPC-21 is not an irreducible fourth-moment obstruction. Pairwise character orthogonality turns it into an arbitrary compatibility matrix tested by two independent character coordinates. Exact conductor resolution then exposes two complementary estimates:

$$Q\sqrt{JY\mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2)}$$

from the inherited norm route, and

$$QF_1F_2\sqrt{\mathcal{K}_Q(F_1)\mathcal{K}_Q(F_2)}$$

from the direct induced-Gauss route. Neither dominates everywhere, but their conductor cells cover the complete selected short–short range in the stated profiles with a fixed power margin.

This closes one concrete analytic gate and removes the arbitrary-fiber obstruction for the actual generic mask in that packet. The remaining route to the full fixed-shift problem still includes the large-divisor legs and global residual reassembly, followed eventually by the classical parity obstruction. Those are separate problems; no claim about them is encoded in the present closure theorem.

References

- [1] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004. doi: 10.1090/coll/053. URL <https://doi.org/10.1090/coll/053>.
- [2] Liang Wang. Assembly and concentration at a fixed-shift type-II gate: Cumulative divisor prefixes, bilinear distribution frontiers, and TT^* dispersion. Preprint, [repository commit 5438204e32f9](#), 2026.
- [3] Liang Wang. Two quadraticizations of a Möbius tail: LCM cutoff return, generic row determinants, and nonprimitive endpoint collapse. Preprint, repository commit [f418ea1a8921](#), 2026.
- [4] Liang Wang. Four-channel compression in a primitive Möbius tail: Centered divisor kernels, a single-leg completion wedge, and CRT-fiber discrepancy. Preprint, [repository commit b545ad5eb0db](#), 2026.
- [5] Liang Wang. Connected fiber energy in a primitive Möbius tail: Ramanujan-mean closure, divisor-lattice reassembly, and a shared-factor character gate. Preprint, [repository commit dc916dc42a2d](#), 2026.